

DP-03 Extractor

Rescue your recordings. Remix them live.

User Guide · Version 0.1.0 · macOS

This guide walks you through recovering audio from the TASCAM DP-03SD's proprietary BFS filesystem using the DP-03 Extractor macOS app. It covers installing the launcher, loading a card, extracting per-track WAV files, and using the built-in live mixer to audition and render stereo mixes.

What's inside

- 1** What This Is & Why It Exists
- 2** System Requirements
- 3** First-Time Setup
- 4** Loading a Card
- 5** The Project List
- 6** Extracting Tracks
- 7** Using the Live Mixer
- 8** Rendering a Stereo Mix
- 9** Troubleshooting
- 10** How It Works (Under the Hood)
- 11** Glossary

1. What This Is & Why It Exists

The TASCAM DP-03SD is a digital portastudio that records eight mono audio tracks per project onto an SD card using an undocumented block filesystem called **BFS**. macOS and Windows both see the card as unformatted. If the recorder fails mid-export or the USB transfer mode glitches, there is no official way to get your recordings back.

DP-03 Extractor parses the BFS filesystem directly off a card image (or the SD card itself) and writes each track as a 44,100 Hz / 16-bit mono **.wav** file. It also ships a real-time stereo mixer so you can audition projects with per-track volume, pan, mute, and solo, then render a stereo bounce.

What this tool does and doesn't do

Does: reads DP-03SD cards, extracts every master per project as a mono WAV, and mixes them down in real time.

Doesn't: edit audio, apply DP-03 effects, read non-TASCAM portastudios, or write back to the SD card. It is strictly read-only against your media.

2. System Requirements

- macOS 10.13 or newer. The app has been tested on macOS 16 (Sequoia) through current releases.
- Python 3.10+ with a working Tkinter. On modern Macs the recommended option is **Homebrew Python 3.12** with the **python-tk@3.12** formula.
- For the live mixer: **numpy** and **sounddevice** installed for the same Python that launches the app.
- A DP-03SD card **or** a full-device image of one (**.img**, **.dmg**). Raw SD access works on macOS if you launch from Terminal with **sudo**.
- About 3–5 GB of free disk for extracted audio, depending on how many songs are on the card.

Apple's /usr/bin/python3 can crash at launch

On some macOS versions Apple ships a Python linked against a newer Tcl/Tk than the OS can actually run, which crashes with “macOS 26 or later required”. The launcher detects this and skips that Python automatically— but you'll need Homebrew's python-tk for the app to start. See Section 3.

3. First-Time Setup

3.1 Install a Tk-capable Python

If you don't already have Homebrew, install it first from **brew.sh**. Then install Python 3.12 with its Tk bindings:

```
brew install python-tk@3.12
```

The launcher will pick up **/opt/homebrew/bin/python3.12** automatically. You can verify it's working by running:

```
/opt/homebrew/bin/python3.12 -c \  
'import tkinter; r = tkinter.Tk(); r.destroy(); print("OK")'
```

If that prints **OK**, Tkinter is ready.

3.2 Install the mixer dependencies

The live mixer needs NumPy and sounddevice. Install them into the same Python the launcher uses:

```
/opt/homebrew/bin/python3.12 -m pip install \  
--user --break-system-packages \  
numpy sounddevice
```

If pip refuses with an “externally-managed-environment” error, the `--break-system-packages` flag above overrides it for a user-level install. If you prefer a venv, create one and symlink it into the launcher search path.

3.3 Install the app

- 1 Copy the **DP-03 Extractor.app** bundle and the adjacent **dp03app** and **dp03extract** folders into the same directory (for example, `~/Applications/dp03`). Keep them side by side— the app bundle resolves the Python package by walking up to its parent folder.
- 2 Double-click **DP-03 Extractor.app**. The first launch may show a Gatekeeper warning; right-click → Open to allow it.
- 3 If the app doesn't appear, run the backup launcher: `Launch DP-03 Extractor.command`. That keeps a Terminal window open so you can read any Python errors.

Launch logs

Every launch writes a short entry to `~/Library/Logs/DP-03 Extractor.log` with the Python it picked and the working directory. If you're stuck, check that file first— it usually tells you which probe succeeded or failed.

4. Loading a Card

4.1 From a disk image

If you've already captured a `.img` or `.dmg` of your DP-03SD card, click **Open card image...** in the toolbar and pick the file. The scan dialog walks the card and lists every project it finds— usually less than 15 seconds for a 16 GB card.

4.2 From the live SD card

Insert the SD card. The app auto-detects TASCAM-formatted media and populates the **Use SD card**  menu. Pick a card from the menu and the scan starts immediately.

Raw device access on macOS

macOS won't let an ordinary user read `/dev/rdiskN` without elevation. If the app shows "Cannot read `/dev/rdiskN` directly," either (a) run the `.` command launcher from a Terminal session that has `sudo` rights, or (b) image the card first with `sudo dd if=/dev/rdiskN of=~/Desktop/card.img bs=4m` and then open that image.

4.3 What you get after a scan

The top strip of the window shows **Card: name** with a count like "189 projects, 153 with audio." The lower project table fills with one row per project. Empty slots (no audio recorded) are dimmed; projects with audio are black.

5. The Project List

Each row in the project list shows:

Column	What it means
#	Slot index on the card (0–249).
Name	8-character project name from the DP-03 display (e.g. SONG116).
Tracks	How many masters contain non-zero audio.
Duration	Length of the longest master in minutes:seconds.

Use -**click** to add or remove individual projects from the selection, or **Shift-click** to select a contiguous range. The row header buttons at the bottom act on your current selection.

6. Extracting Tracks

Pick an output folder below the project list (default: `~/Desktop/DP-03 Extracts`), then decide what to extract.

- **Extract selected** — writes one subfolder per highlighted project.
- **Extract all with audio** — processes every project where Tracks > 0. On a full card this takes several minutes and produces a few GB of WAVs.
- **Cancel** — stops the queue after the current project finishes.

6.1 Output layout

```
DP-03 Extracts/  
116_SONG116/  
Track1.wav  
Track2.wav  
Track3.wav  
...  
141_SONG141/  
Track1.wav  
Track2.wav
```

Each WAV is a standalone 16-bit / 44.1 kHz mono file, byte-identical to what the DP-03SD writes when you run its built-in “EXPORT TRACK” feature. Only masters with declared non-zero length are written; empty slots are skipped.

Reveal output folder

Click **Reveal output folder** to jump to the destination in Finder. The folder is created automatically if it doesn't exist yet.

7. Using the Live Mixer

Select **one** project with audio, then click **Open mixer...**. The app decodes every master in the background and opens a mixer window with a channel strip per track. Two-track projects are auto-panned L/R for a natural stereo feel; everything else starts at center.

7.1 The channel strip

- **Fader** — Vertical slider from $-\infty$ dB (bottom) to +6 dB (top). The dB value is shown underneath. Moving the fader to the very bottom mutes the track completely.
- **Pan** — Horizontal slider from L100 to R100. Equal-power panning keeps perceived loudness constant across the stereo field.
- **Mute** — Removes the track from the mix without changing its fader.
- **Solo** — When any track is soloed, all non-soloed tracks are silenced. Solo wins over Mute.

7.2 Transport bar

- **■ Play** — Start playback from the current position. The button label changes to “Playing” while audio is streaming.
- **■ Pause** — Halts the stream; press Play to resume where you stopped.
- **■ Stop** — Stops and rewinds to 0:00.
- **Time readout** — Shows *current / total* in minutes:seconds.hundredths.
- **Scrub bar** — Drag to seek anywhere in the project. The transport auto-follows when you release the mouse, so you can slam to any point mid-playback without glitching.

Solo vs. Mute precedence

If you solo Track 2 while Track 1 is muted, only Track 2 plays. Un-solo Track 2 and Track 1 stays muted until you un-mute it. That matches the behavior of most DAWs.

8. Rendering a Stereo Mix

Once a mix sounds right, click **Render Mix...** in the mixer’s transport bar. Pick a destination file and the app will freeze the current state of every fader, pan pot, mute, and solo into a new stereo 16-bit WAV.

- The render is offline — audio is not streamed through your output device, and the result is identical regardless of your current sample-rate / buffer-size settings.
- Any track at $-\infty$ dB or muted (and not soloed) is skipped entirely; the output is all-zero samples where no tracks contribute.
- The final mix is clipped to $[-1.0, +1.0]$ before being quantized to int16, matching the behavior of the live output.
- A progress bar shows percent complete; a 4-minute project usually renders in under 2 seconds.

Where renders land

The default save location is the output folder you picked in the main window. You can override it per-render from the save dialog. After the render finishes you're offered a "Reveal in Finder" shortcut.

9. Troubleshooting

Symptom	What to do
App bounces once in the Dock, then nothing	Launch from Launch DP-03 Extractor.command instead. It keeps the Terminal open so you can read the error. Usually means no Tk-capable Python was found.
"macOS 26 or later required" in the log	Apple's stock Python has a broken Tk for your macOS. Run <code>brew install python-tk@3.12</code> and re-launch.
"ImportError: No module named _tkinter"	Homebrew's default python3 builds without Tk. Install <code>python-tk@3.12</code> or use Apple's <code>/usr/bin/python3</code> .
Open mixer... is greyed out	Select exactly one project that has at least one track. The button is disabled for empty projects and multi-selects.
Mixer popup: "needs numpy and sounddevice"	Run the pip install from Section 3.2 using the same Python the launcher uses (check the log).
No audio from the mixer	Make sure macOS's output device is unmuted and the volume is up. sounddevice uses the default CoreAudio output; change it in System Settings → Sound.
Scan finds zero projects on a known-good card	Confirm you're pointed at a DP-03SD image, not a FAT32 extract. BFS lives outside the FAT partition— you need a raw dd image of the whole card.
Render Mix produces silence	Check that at least one track has a fader above $-\infty$ and is not muted (and, if any track is soloed, that it is the soloed one).

10. How It Works (Under the Hood)

This section is optional reading for anyone curious about the BFS recovery pipeline.

10.1 Finding projects

DP-03SD cards have a proprietary “MTR” region outside the FAT32 partition. Project roots are identified by a fixed anchor pattern at base + 0x40:

```
00 00 00 40 80 00 00 00 xx xx xx xx E3 1B A5 FF
```

Followed by an 8-character ASCII name at base + 0x50. The scanner walks the whole card looking for this signature and builds a ProjectSummary for each hit.

10.2 Walking the extent tree

Each project's allocation table points to master records; each master's +0x24 field points to a Song File Descriptor, which itself points (at +0x14) to an inode. The inode's +0x24 field is the head of an extent-leaf linked list. Each leaf holds up to six (*cluster_id*, *byte_offset*) pairs; leaf-to-leaf chaining is via the leaf's +0x0C pointer.

10.3 Re-assembling audio

Once every (*cluster_id*, *byte_offset*) pair is collected, the extractor sorts them by *byte_offset* (time order), reads the full 96 KiB sector at *mtr_base* + *cluster_id* × 0x18000, concatenates them, and truncates to *samples* × 2 bytes. The result is byte-identical to the DP-03's built-in TEX001 WAV exports.

10.4 The mixer engine

Extracted PCM is wrapped in a TrackState (an int16 ndarray plus *volume_dB*, *pan*, *mute*, *solo*). A sounddevice OutputStream at 44,100 Hz / 2 channels / float32 / 1024-frame blocks pumps audio. Each callback snapshots state under a short lock, converts slices to float32, applies $\text{gain} = 10^{(\text{dB}/20)}$ and equal-power pan ($\text{angle} = (\text{pan}+1) \cdot \pi/4$; $L=\cos$; $R=\sin$), mixes into the output buffer, and clips to [-1, +1]. Render Mix runs the same math in 4-second chunks without touching CoreAudio.

11. Glossary

BFS	TASCAM's proprietary Block File System used on the portion of the SD card outside the FAT32 partition.
Master	One DP-03 track's audio stream plus metadata.
Sector / cluster	A 0x18000-byte (96 KiB) unit of audio storage on the card.
Extent leaf	A record with up to six (<i>cluster_id</i> , <i>byte_offset</i>) pairs describing how to reassemble a master.
MTR region	The proprietary area of the card used for BFS. FAT tooling will not see it.
TEX001	The folder the DP-03 writes mono WAV exports to when you use its on-device EXPORT TRACK feature. Extractor output matches these byte-for-byte.
Equal-power pan	A pan law that keeps perceived loudness constant across the stereo field; center-panned signals are –3 dB relative to full L/R.

Version 0.1.0 · Built for the 16 GB TASCAM DP-03SD cards imaged in April 2026. Extraction has been validated byte-identical against the DP-03's own TEX001 exports.